

1 MBL Range of Coverage Analysis

Overview

The range of coverage analysis allows the user to recognize how much time is left until a specific material for a production level is consumed and/or for how long there will be enough material available for production.

This means that this function analyzes the range of coverage of a material transferred between two consecutive production levels (preceding level and subsequent level).

The range of coverage computation is requested for a *production level* considered as consuming in this context. With regard to the analysis, all supply levels/production levels assigned to this level as preceding levels are then considered.

If these, in turn, are supply levels to other levels, the latter are also included for the purpose of a holistic analysis.

In this process, the following materials from the batch stock are included:

- Material of the selected production level
 - Material in the configured input buffer of the production level
 - Material currently being used as component in the running operation (to be found through BOM)
- Material from the preceding levels to the selected production level
 - Material currently located in configured output buffers of preceding levels
 - Material currently produced in the preceding levels of the running operation

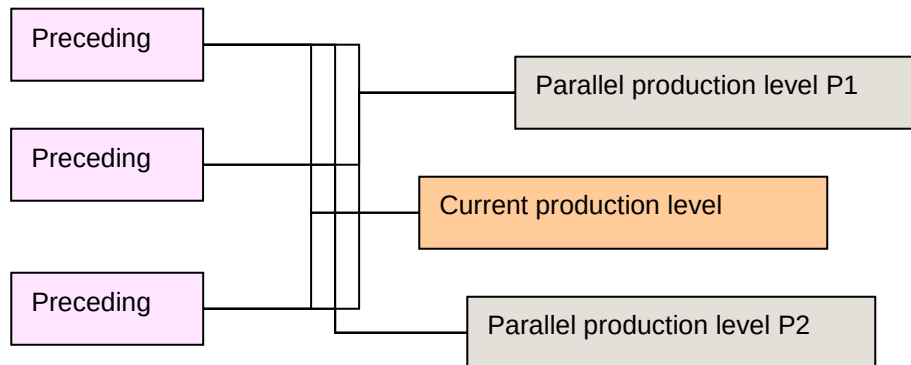
For each material, the number of supplying as well as consuming machines is indicated.

The current stock of material buffers to be analyzed is considered as initial stock. In this regard, batches currently logged on to consuming machines are also considered.

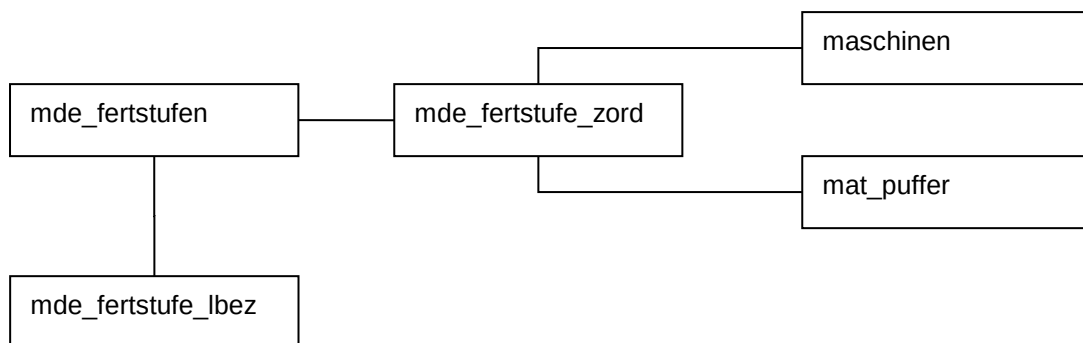
For calculating the range of coverage, the criteria listed below are therefore relevant:

- The consuming OPs and their target cycle result in a demand per unit of time (e.g. component A = 100 pcs/h are consumed).
- The supplying OPs and their target cycle result in a stock per unit of time (e.g. component A = 100 pcs/h are produced).

Connections



Configuration Data Model



Selection

- Production level
The range of coverage computation is requested for a production level considered as consuming in this context. With regard to the analysis, all production levels assigned to this level as preceding levels are then considered. If these, in turn, are supply levels to other levels, these other, retrieving production levels are also included for the purpose of a holistic analysis.
- Incl. materials which are (currently) only supplied/produced
This option allows for extending the analysis to also include materials which are currently only supplied and/or only produced.

Display

The range of coverage is displayed in hours as well as a trend (arrows/signs/symbols). The display is updated automatically and/or cyclically every 180 seconds.

The range of coverage analysis grid shows the data listed below:

Designation	Description, source
Material	
Material	- In producing operations, the material corresponds to the materials in the component list - In supplying operations, the material corresponds to the article ID
Designation	Material designation from batch stock for selected material numbers
Stock	Quantity: - Batches to be selected from batch stock with relevant material number - Selected batches are always in the free/running status (in buffer/posted on operation). Processed or locked batches are not used. - If the remaining quantity is < 0, these batches are not considered.
Unit	Unit from batch stock for selected quantity
Range of coverage	
Range of coverage	Range of coverage in hours (decimal); default: not visible
Previous range of coverage	Range of coverage in hours (decimal); default: not visible
Trend	In addition, the trend of the range of coverage (change in comparison to previous state) is indicated in front of each bar by an appropriate symbol. - increasing  - decreasing  - unchanged 
1	Hour; bar chart
:	Hour; bar chart
16	Hour; bar chart
17	Hour, bar chart; default: not visible
:	Hour, bar chart; default: not visible
24	Hour, bar chart; default: not visible
Retrieving machines	
Total	Machines within the supply relationships of the current production level and with currently running operations using this material.
In production	Machines within the supply relationships of the current production level and with currently running operations using this material and operating in the <i>Production</i> status.
Supplying machines	
Total	Machines of other supply relationships with currently running operations using this material.
In production	Machines within the supply relationships of the current production level and with currently running operations producing this material.

Data Acquisition

Data are acquired for a production level considered as consuming in this context. With regard to the analysis, all production levels assigned to this level as preceding levels are then considered. If these, in turn, are supply levels to other levels, these (other levels) are also included for the purpose of a holistic analysis.

Parameter Entry

FSTUFE : selected production level

PROD=J : incl. materials (currently) only supplied/produced
→ no range of coverage is calculated for these (range of coverage = 0)

Value Return

Column	Comment/Layout
ARTIKEL	Material
BEZ	Material designation
BESTAND	Calculated material stock
EINHEIT	Quantity unit of material
REICHW	Range of coverage: calculated value
ANZ_ENT_MNR	Number of retrieving machines
ANZ_ENT_MNR_PROD	Number of retrieving machines in production
ANT_ENT_MNR_ANDERE	Number of retrieving machines from other production levels
ANZ_PROD_MNR	Number of producing machines
ANZ_PROD_MNR_PROD	Number of producing machines in production
MIN_REICHW	Minimum range of coverage of material; always 0!

The following criteria are relevant for the range of coverage:

- The consuming OPs (current production level and production levels running in parallel, if any) and their cycle result in a demand per unit of time (e.g. component A = 100 pcs/h are consumed).
- The supplying OPs (preceding production levels) and their cycle result in a stock per unit of time (e.g. component A = 100 pcs/h are produced).

The initial stock is the current material stock from assigned material buffers of preceding levels. In this regard, batches currently logged on to consuming machines must also be considered.

Procedure

Identification of all machines to be considered

Recommendation: Enter the machines in a temporary table.

- Machines of the current production level

- Machines of production levels preceding the current production level (preceding levels)
- Machines of other production levels following the preceding levels (quasi existing in parallel 'P' to the current production level)

Identification of materials used on machines of the current level

(For these, the range of coverage has to be determined, i.e. how long will the current stock last.)

For each currently running operation (AUFTRAG_STATUS.PROD_KENN = 'L') and its consuming material components (MLST_HY.KENNZ 'M', 'T') in production on machines of the current production level, the required quantity per hour is calculated.

The basis for this is the target cycle entered for the operation (= speed at which the article is produced; AUFTRAGS_BESTAND.SOLL_DAUER/SOLL_TEIL) as well as the input quantity of the supplied material components (MLST_HY.SOLL_MENGE).

$$\text{Time required for producing 1 piece} = \frac{ab.soll_dauer / ab.soll_teil}{1000} \text{ [s]}$$

$$\text{Time required for a material to be used} = \text{Timeperpcs} = \frac{\left(\frac{ab.soll_dauer / ab.soll_teil}{1000} \right)}{mlst_hy.soll_menge} \left[\frac{s}{UOM} \right]$$

$$\text{Required quantity per hour for a material to be used} = \frac{3600 \text{ [s]}}{\text{TimePerItem} \left[\frac{s}{UOM} \right]}$$

Since an article may be processed at different target cycle times in different orders/operations, the individual values are determined as an added up total.

When calculating the range of coverage, the OPs logged on to machines of levels running in parallel (P1, P2) must also be considered. For this reason, the "Required quantity per hour for a material to be used" must be identified for each of them, too.

Identification of initial stocks

The initial stock is the current material stock (LOS_BESTAND.RESTMENGE) from assigned material buffers of preceding levels. The batches with "L" and "F" status (LOS_BESTAND.STATUS) whose residual quantity is > 0 are considered.

Identification of materials produced on machines of preceding levels

For materials produced in the preceding level, the speed at which the current operations (AUFTRAG_STATUS.PROD_KENN = 'L') produce the materials on machines of preceding levels (AUFTRAGS_BESTAND.ARTIKEL) is calculated.

$$\text{Time required for producing 1 piece} = \text{Timeperpcs} = \frac{\text{ab.soll_dauer} / \text{ab.soll_teil}}{1000} \text{ [s]}$$

$$\text{Quantity produced per hour} = \frac{3600 \text{ [s]}}{\text{TimePerItem} \left[\frac{\text{s}}{\text{UOM}} \right]}$$

Calculation of range of coverage

The range of coverage of each material [in hours] is determined from the ratio of the values described above:

$$\text{Range of coverage [h]} = \frac{\text{LOS_BESTAND.RESTMENGE}}{\text{TimePerItem (ANR1)} + \text{TimePerItem (ANR2)} + \dots}$$

$$\frac{3600.0}{\text{TimePerItem (ANR1)}} + \frac{3600.0}{\text{TimePerItem (ANR2)}} - \frac{3600.0}{\text{TimePerItem (ANRn)}} + \dots$$

+ producing OPs (range of coverage increases)

- consuming OPs of current level (range of coverage decreases)

- consuming OPs of level(s) running in parallel (range of coverage decreases even more)

If the option "Incl. materials which are (currently) only supplied/produced" is set, not only the machines used in the current production level but also the materials only produced in the preceding level, are identified. For these, however, only the stock, but not the range of coverage will be calculated.

Number of retrieving machines

Number of retrieving machines (i.e. of the production level to be considered) currently using this article (based on component list of active order).

Number of retrieving machines (i.e. of the production level to be considered) currently in PRODUCTION (STOER_TABELLE.PROD_KENN = 'P') and using this article (based on component list of active order).

Number of retrieving machines from other production levels

Number of retrieving machines in production levels which, in relation to preceding levels, are downstream and which currently use the articles (based on component list of active order) also used on machines of the current production level.

Number of supplying machines

Number of supplying machines (i.e. of the preceding levels) currently producing these articles.

Number of supplying machines (i.e. of the preceding levels) currently producing these articles and in PRODUCTION.